Brac University

Department of Electrical & Electronic Engineering

Semester Spring-25

Course Number: EEE103L

Course Title: Computer Programming Laboratory

Section: 06

Lab Report

Experiment no. 03 (Part 2)

Name of the experiment: Familiarization with Decision Control Structures & Loop Control Structures.

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Group Number: 04

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Lab report of experiment 3 (Part 2)

C. Write a program for a matchstick game being played between the computer and a user. Your program should ensure that the computer always wins. Rules for the game are as follows:

- ☐ There are 21 matchsticks.
- ☐ The computer asks the player to pick 1, 2, 3, or 4 matchsticks.
- ☐ After the person picks, the computer does its picking.
- ☐ Whoever is forced to pick up the last matchstick loses the game.

<u>Code:</u>

```
#include <stdio.h>

int main()
 {
   int stick = 21, player, computer;

   printf("Starting the matchstick game\n");
   printf("Rules: Pick 1, 2, 3, or 4 matchsticks on each turn. Whoever picks the last
matchstick will lose.\n\n");

   while (stick > 1) {
      printf("Matchsticks left: %d\n", stick);
      printf("Pick 1, 2, 3, or 4 : ");
      scanf("%d", &player);

      if (player < 1 || player> 4)
         {
         printf("Wrong count. Pick between 1 to 4.\n");
      }

      stick = stick - player;

      if (stick == 1)
      {
         printf("Last Matchstick picked. You lose :D \n");
      }

      computer= 5 - player;
      printf("Computer picks %d matchstick.\n", computer);

     stick = stick - computer;
```

```
        if (stick == 1)
          {
          printf("Forced to pick the last one. You lose :D\n");
        }
    }
    return 0;
}
```

Screenshot of code:

matchstick.c

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```c
#include <stdio.h>
int main()
 {
    int stick = 21, player, computer;

    printf("Starting the matchstick game\n");
    printf("Rules: Pick 1, 2, 3, or 4 matchsticks on each turn. Whoever picks the last matchstick will lose.\n\n");

    while (stick > 1) {
        printf("Matchsticks left: %d\n", stick);
        printf("Pick 1, 2, 3, or 4 : ");
        scanf("%d", &player);

        if (player < 1 || player> 4)
            {
            printf("Wrong count. Pick between 1 to 4.\n");
        }

        stick = stick - player;

        if (stick == 1)
        {
            printf("Last Matchstick picked. You lose :D \n");
        }

        computer= 5 - player;
        printf("Computer picks %d matchstick.\n", computer);

       stick = stick - computer;

        if (stick == 1)
            {
            printf("Forced to pick the last one. You lose :D\n");
        }
    }
    return 0;
}
```

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Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\aryan>  & 'c:\Users\aryan\.vscode\extensions\ms-vscode.cpptools-1.23.6-win32-x64\debu
in=Microsoft-MIEngine-In-jvhvwznu.bzu' '--stdout=Microsoft-MIEngine-Out-aans1yyo.2vq' '--stderr=M
icrosoft-MIEngine-Pid-0adzyd3z.rb3' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
Starting the matchstick game
Rules: Pick 1, 2, 3, or 4 matchsticks on each turn. Whoever picks the last matchstick will lose.

Matchsticks left: 21
Pick 1, 2, 3, or 4 : 3
Computer picks 2 matchstick.
Matchsticks left: 16
Pick 1, 2, 3, or 4 : 2
Computer picks 3 matchstick.
Matchsticks left: 11
Pick 1, 2, 3, or 4 : 3
Computer picks 2 matchstick.
Matchsticks left: 6
Pick 1, 2, 3, or 4 : 4
Computer picks 1 matchstick.
Forced to pick the last one. You lose :D
```

d. Write a program to produce the following output:

```
      1
     2 3
   4  5  6
         7  8 9 10
```

Code:

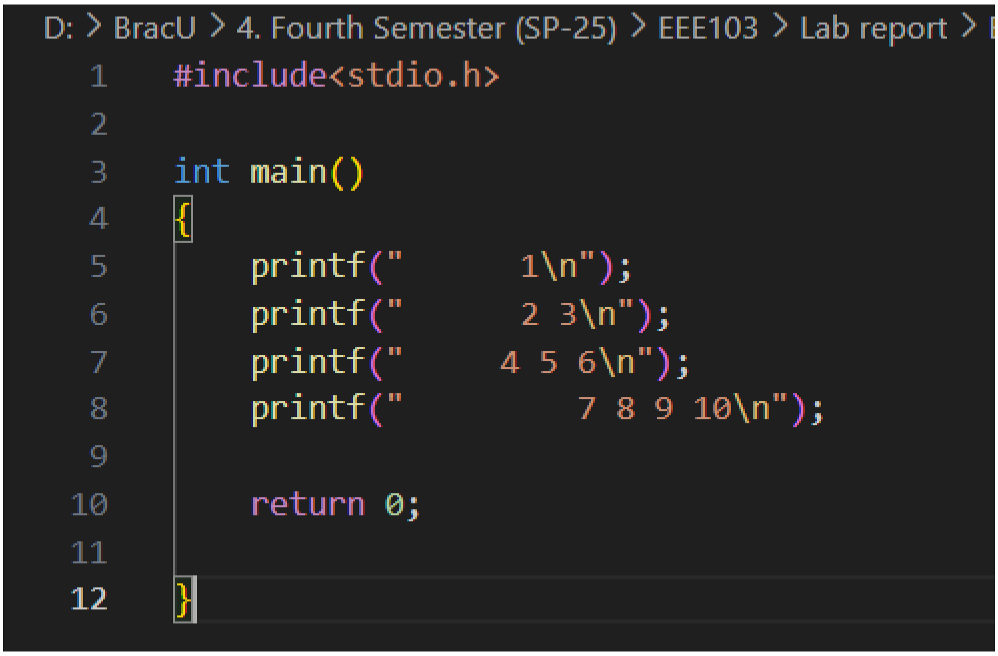
```
#include<stdio.h>

int main()
{
    printf("     1\n");
    printf("     2 3\n");
    printf("    4 5 6\n");
    printf("        7 8 9 10\n");

    return 0;

}
```

Screenshot of code:



```
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#include<stdio.h>

int main()
{
    printf("      1\n");
    printf("      2 3\n");
    printf("     4 5 6\n");
    printf("         7 8 9 10\n");

    return 0;

}
```
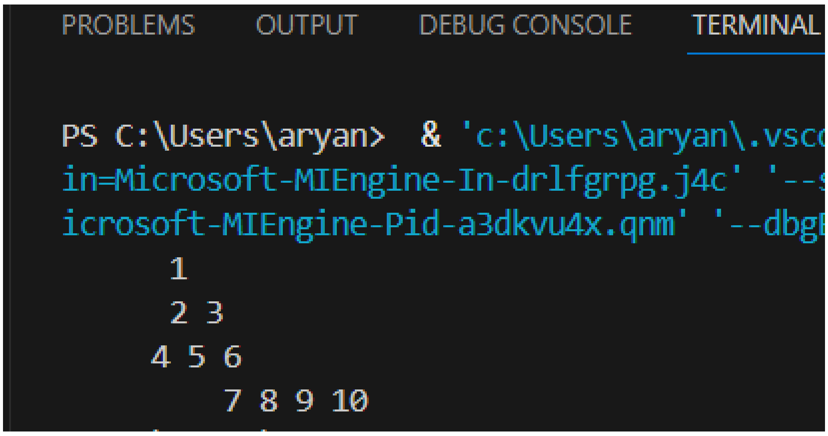

Output:



```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL

PS C:\Users\aryan>  & 'c:\Users\aryan\.vsc
in=Microsoft-MIEngine-In-drlfgrpg.j4c' '--
icrosoft-MIEngine-Pid-a3dkvu4x.qnm' '--dbg
      1
      2 3
     4 5 6
         7 8 9 10
```

e. Population of a town today is 100000. The population has increased steadily at the rate of 10 % per year for last 10 years. Write a program to determine the population at the end of each year in the last decade.

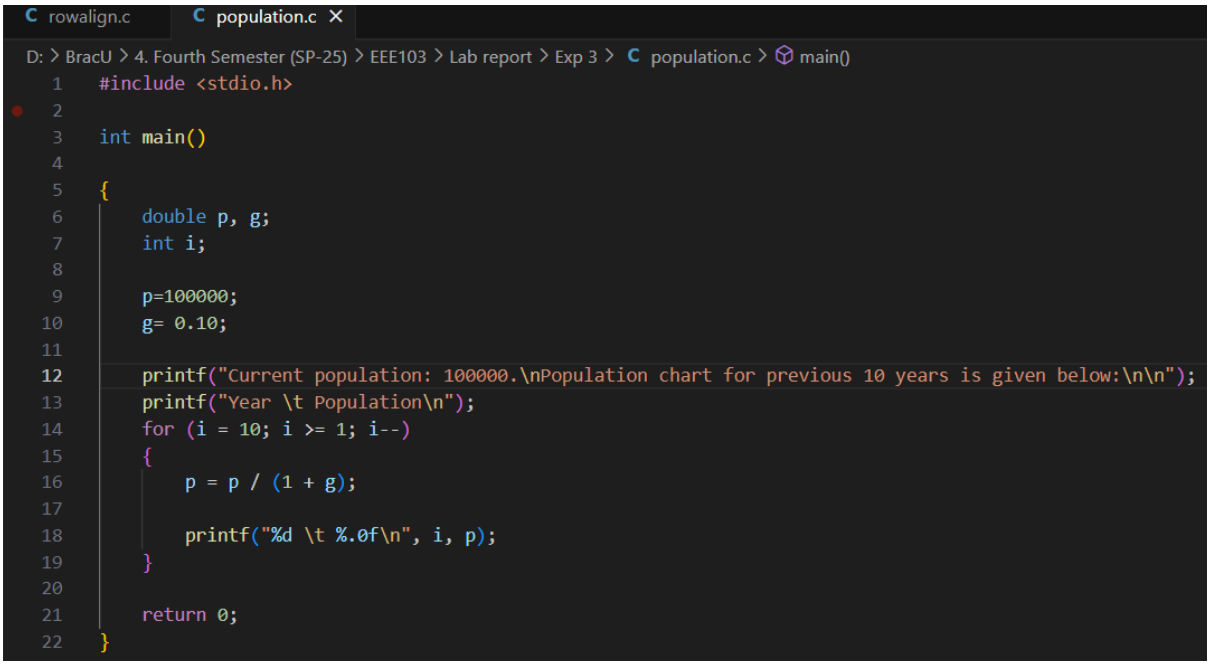
Code:

#include <stdio.h>

int main()

```
{
    double p, g;
    int i;

    p=100000;
    g= 0.10;

    printf("Current population: 100000.\nPopulation chart for previous 10 years is given below:\n\n");
    printf("Year \t Population\n");
    for (i = 10; i >= 1; i--)
    {
        p = p / (1 + g);

        printf("%d \t %.0f\n", i, p);
    }

    return 0;
}
```

<u>Screenshot of code:</u>



Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\aryan>  & 'c:\Users\aryan\.vscode\extensions
in=Microsoft-MIEngine-In-flve1wly.u2g' '--stdout=Microso
icrosoft-MIEngine-Pid-rtb0zc25.gyi' '--dbgExe=C:\msys64\
Current population: 100000.
Population chart for previous 10 years is given below:

Year      Population
10        90909
9         82645
8         75131
7         68301
6         62092
5         56447
4         51316
3         46651
2         42410
1         38554
```